Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2025-P1-Model)** **Date: 29-03-2026**

Time: 09.00Am to 12.00Pm **GTA-21** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**C**	2	**C**	3	**D**	4	**C**	5	**ABC**	6	**ACD**
7	**BC**	8	**36**	9	**3**	10	**7**	11	**40**	12	**77**
13	**2**	14	**D**	15	**C**	16	**D**				

PHYSICS

17	**D**	18	**D**	19	**A**	20	**B**	21	**BC**	22	**ACD**
23	**ACD**	24	**1.50**	25	**1**	26	**45**	27	**30**	28	**4**
29	**96**	30	**C**	31	**C**	32	**A**				

CHEMISTRY

33	**A**	34	**B**	35	**A**	36	**D**	37	**AB**	38	**BCD**
39	**ABD**	40	**35**	41	**4**	42	**113**	43	**1**	44	**10**
45	**2**	46	**A**	47	**C**	48	**D**				

For More Join: @IITJEE_Advanced

SOLUTIONS
MATHEMATICS

1. $Q^{-1}=Q^T \Rightarrow QQ^T=I$

Q is orthogonal matrix

Let $Q=\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$

$$PQ=QP \Rightarrow \begin{bmatrix} 2a_1 & 2b_1 & 2c_1 \\ 2a_2 & 2b_2 & 2c_2 \\ 3a_3 & 3b_3 & 3b_3 \end{bmatrix}=\begin{bmatrix} 2a_1 & 2b_1 & 3c_1 \\ 2a_2 & 2b_2 & 3c_2 \\ 2a_3 & 2b_3 & 3c_3 \end{bmatrix}$$

$c_1=0, c_2=0, a_3=0, b_3=0$

$Q=\begin{bmatrix} a_1 & b_1 & 0 \\ a_2 & b_2 & 0 \\ 0 & b_3 & c_3 \end{bmatrix}$ $a_1a_2+b_1b_2=0$

$a_1^{\,2}+b_1^{\,2}=1, \quad a_2^{\,2}+b_2^{\,2}=1, \quad c_3^{\,2}=1$

a_1	b_1	a_2	b_2	c_3
1	0	0	$1,-1$	$+1,-1$
-1	0	0	$1,-1$	$1,-1$
0	1	$1,-1$	0	$1,-1$
0	-1	$1,-1$	0	$1,-1$

Total 16 matrices

2. To solve the equation, a substitution like $t=\tan\theta$ can be used. Further substitution, such as $t=3\tan\theta$, simplifies the expression within the $\sin^{-1}$ function. The equation then involves terms like $\sin^{-1}(\sin(2\phi))$. Considering the range of 2ϕ and the principal value branch of $\sin^{-1}x$, different cases arise. Solving these cases leads to solutions for $\tan\theta$, which are found to be $0, 1,$ and -1. These values correspond to $\theta=0, \frac{\pi}{4},$ and $-\frac{\pi}{4}$ within the principal value branch of $\tan^{-1}\theta$, resulting in 3 distinct solutions.

3. $a=0, b=4$ check graphically

4. $\vec{c}\times\vec{a}=\vec{b}$

$\Rightarrow |\vec{b}|=|\vec{c}||\vec{a}|\sin\alpha$

$\Rightarrow |\vec{c}|=\frac{3}{2}\operatorname{cosec}\alpha.$

$|\vec{c}-\vec{a}|^2=\frac{9}{4}\operatorname{cosec}^2\alpha+4-2.2.\frac{3}{2}\operatorname{cosec}\alpha\cdot\cos\alpha$

For More Join: @IITJEE_Advanced

$$=\frac{9}{4}+\left(\frac{3}{2}\cot\alpha-2\right)^2\geq\frac{9}{4}$$

So clearly $|\vec{c}-\vec{a}|_{\min}=\frac{3}{2}@\tan\alpha=\frac{3}{4}$

$$|\vec{c}|=\frac{3}{2}\cdot\frac{5}{3}=\frac{5}{2}$$

5. Conceptual

6. $S(n,k)\equiv$No. of ways to partition n distinct items into k parts.

First consider one particular element a_1 forming one group and remaining $n-1$ elements forming $k-1$ partitions.

Second, consider $n-1$ elements forming k partitions and a, element going in one 'k' groups.

Hence, $S(n,k)=S(n-1,k-1)+kS(n-1,k)$

No. of onto functions $=k!S(n,k)$

$$=\left[k^n-{}^kC_1(k-1)^n+{}^kC_2(k-2)^n.....\right]$$

$$S(5,3)=\frac{1}{3!}\left[3^5-{}^5C_1 2^5+{}^3C_2 1^5\right]=25$$

No. of onto functions from 6 elements $=3!S(6,3)$ to 3 elements.

7. Clearly $S\equiv(4,0)$; Let $P=(\alpha,\beta)$

$\Rightarrow a+e\alpha=7\Rightarrow 5+\frac{4}{5}\alpha=7\Rightarrow\alpha=\frac{5}{2}$ and $\beta=\frac{3\sqrt{3}}{2}$

Hence, equation of tangent at P is $\sqrt{3}x+5y-10\sqrt{3}=0$

As tangent is bisector of angle between SP and $S'P$

$\Rightarrow PQ=PS=3$. As $PS+PS'=10=$Length of major axis when P is variable then

$S'Q=S'P=PQ=S'P+PS=10$

$\Rightarrow$Variable point Q is at a fixed distance from S'

$\Rightarrow$Locus is a circle with centre S' and radius = 10 units

8. $f(x)+g(x)=a(x+1)(x+2)(x+3)$

$f(x)-g(x)=b(x-1)(x-2)(x-3)$

So $f(0)=3(a-b),g(0)=3(a+b)$

Hence $f^2(0)+g^2(0)=18\left(a^2+b^2\right)\geq 36$ (equality holds for $a,b\in\{-1,1\}$)

9. For every acute angled triangle there are 3 obtuse angled triangles.

10. $\left({}^{19}c_0\left(x^3\right)^{19}+{}^{19}C_1\left(px^2+2x-5\right)\left(x^3\right)^{18}+....\right)\left({}^8C_0x^{16}+{}^8C_1(qx-41)x^{14}+...\right)$

$$\left({}^6C_0x^{24}+{}^6C_1\left(-x^3+x-1\right)x^{20}+...\right)=x^{97}+391x^{96}=a_{95}x^{95}+...$$

Comparing the coefficient of x^{96} we get $19p+8q-6=391$

Rankers Academy JEE
For More Join: @IITJEE_Advanced

$\Rightarrow 19p+8q=397$ Let $q=19\lambda+k, 0\le k\le 19$

$$p=\frac{397-8(19\lambda+k)}{19}=21-8\lambda-2\frac{(4k+1)}{19}$$

$\frac{4k+1}{19}$ must be integer $\Rightarrow k=14, p=15,-8\lambda$

For minimum positive value of $p, \lambda=1\Rightarrow p=7$

11. $$P(n)=\left(\frac{2^n+(1+w)^n+\left(1+w^2\right)^n-3}{3.2^n}\right)/\left(1-\frac{1}{2^n}\right)$$

$$=\frac{2^n-3+(-1)^{n+1}}{3\left(2^n-1\right)}$$

12. $\sqrt{x}=t$

$$y=t^3+\frac{1}{8t^3}-\frac{17}{4}\left(t^2+\frac{1}{4t^2}\right)$$

$$y=\left(t+\frac{1}{2t}\right)^3-3\left(\frac{t}{2}+\frac{1}{4t}\right)-\frac{17}{4}\left(\left(t+\frac{1}{2t}\right)^2-1\right)$$

$y=u^3-\frac{3}{2}(u)-\frac{17}{4}\left(u^2-1\right)$ $\left(\text{where} u=t+\frac{1}{2t}\ge 2\right)$ which has minimum at $u=3$

$$y(3)=27-\frac{9}{2}-\frac{14}{7}(8)=-\frac{23}{2}$$

13. $a_{n+1}+b_{n+1}=2\left(a_n+b_n\right)$

So $a_n+b_n=-2^n$

Also $a_{n+1}\cdot b_{n+1}=2a_nb_n$

So $a_n\cdot b_n=-2^{n+1}$

a_{2012} is bigger root of $x^2+2^{2012}x-2^{2013}=0$

Hence $a_{2012}=-2^{2011}+2^{1006}\sqrt{2^{2010}+2}; m=1006, n=2010$

14. A) $$\frac{\left(\frac{49a}{3}\right)3+\left(\frac{3b}{2}\right)2+c}{6}\ge\left(\left(\frac{49a}{3}\right)^3+\left(\frac{3b}{2}\right)^2+c\right)^{1/6}=\left(\frac{7^2a^3b^2c}{12}\right)^{1/6}$$

$\Rightarrow 49a+3b+c\ge 6\times 7=42$

B) $x\equiv -t, t>0$

$$2x^3-\frac{3}{x^2}=-\left(2t^3+\frac{3}{t^2}\right)\le -5$$

For More Join: @IITJEE_Advanced

$$\because \frac{t^3+t^3+\frac{1}{t^2}+\frac{1}{t^2}+\frac{1}{t^2}}{5} \geq \left(\left(t^3\right)^2\frac{1}{\left(t^3\right)^2}\right)^{1/5} = 1 \Rightarrow 2t^3+\frac{3}{t^2} \geq 5$$

C) $$\left(\frac{\frac{x^3}{5}+\frac{x^3}{5}+\frac{x^3}{5}+\frac{x^3}{5}+\frac{x^3}{5}+\frac{8-x^3}{3}}{8}\right)^8 \geq \left(\frac{8-x^3}{3}\right)\left(\frac{x^3}{5}\right)$$

$$\Rightarrow \left(8-x^3\right)x^5 \leq \left(3^3 5^5\right)^{1/3} = 3\times 5^{5/3}$$

D) $$\frac{(x+x+x+x+x+x+x)+(y+y+y+y+y)}{12} \geq \left(x^7y^5\right)^{1/12} \Rightarrow (7x+5y) \geq 12a^{1/2}$$

$$\therefore 12a^{1/12} \geq 12 \Rightarrow a \geq 1$$

15. $$\frac{f(x)}{x} = \begin{vmatrix} (\alpha x+1)\cos^2 x & 1 & 1-x \\ \beta \sin x & x & 2x \\ (\gamma x^2+1)\tan x & 1 & 1-x^2 \end{vmatrix} \Rightarrow \frac{f(x)}{x^2} = \begin{vmatrix} (\alpha x+1)\cos^2 x & 1 & 1-x \\ \frac{\beta \sin x}{x} & 1 & 2 \\ (\gamma x^2+1)\tan x & 1 & 1-x^2 \end{vmatrix}$$

$$\lim_{x\to 0}\frac{f(x)}{x} = 0, \lim_{x\to 0}\frac{f(x)}{x^2} = \begin{vmatrix} 1 & 1 & 1 \\ \beta & 1 & 2 \\ 1 & 1 & 1 \end{vmatrix} = -1$$

$$f(0) = f'(0) \Rightarrow \lim_{x\to 0}\frac{f'(x)}{x}\lim_{x\to 0} f''(x) = f''(0)$$

$$f''(0) = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 & -1 \\ 0 & 0 & 2 \\ 0 & 1 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 1 & -1 \\ 0 & 0 & 2 \\ 0 & 1 & 0 \end{vmatrix} = -2$$

$$A = \lim_{x\to 0}\frac{1}{x^6}\int_{x^2}^{x^3} f(x)\,dx = \lim_{x\to 0}\frac{f\left(x^3\right).3x^2 - f\left(x^2\right).2x}{6x^5}$$

$$= \lim_{x\to 0}\frac{3xf\left(x^3\right)-2f\left(x^2\right)}{6x^4} = \lim_{x\to 0}\frac{3f\left(x^3\right)+9x^3f'\left(x^3\right)-4xf'\left(x^2\right)}{24x^3}$$

$$= \lim_{x\to 0}\frac{3f\left(x^3\right)-4xf'\left(x^2\right)}{24x^4}+\frac{9}{24}f'\left(x^3\right) = \lim_{x\to 0}\frac{9x^2f'\left(x^3\right)+4f'\left(x^2\right)-8x^2f''\left(x^2\right)}{72x^2}+\frac{9}{24}f'\left(x^3\right)$$

For More Join: @IITJEE_Advanced

$$= \lim_{x\to 0} \frac{-4f'\left(x^2\right)-8x^2 f''\left(x^2\right)}{72x^2} + \frac{2}{2}f'\left(x^3\right)$$

$$= \lim_{x\to 0} \frac{-8x^2 f''\left(x^2\right)-16x f''\left(x^2\right)-16x^3 f'''\left(x^2\right)}{144x} + \frac{1}{2}f'\left(x^3\right) = \frac{48}{144} = \frac{1}{3} \Rightarrow [A] = 0$$

$$g(x) = \begin{vmatrix} \cos^2 x & 1 & 1-x \\ 0 & 1 & 2 \\ \tan x & 1 & 1-x^2 \end{vmatrix} = \cos^2 x\left(1-x^2-2\right) + \tan\left(2-1+x\right)$$

$$= -\cos^2 x\left(1+x^2\right) + \tan x(1+x)$$

$$g\left(\frac{\pi}{4}\right) = -\frac{1}{2}\left(1+\left(\frac{\pi}{4}\right)^2\right) + 1 + \frac{\pi}{4} = \frac{1}{2} + \frac{\pi}{4} - \frac{1}{2}\left(\frac{\pi}{4}\right)^2 = 0.97\ (Approx.)\ \left[g\left(\frac{\pi}{4}\right)\right] = 0$$

16. A) As we know, $|z_1 - z_2|^2 = |z_1|^2 + |z_2|^2 - 2\,\mathrm{Re}\, z_1 - \bar{z}_2$

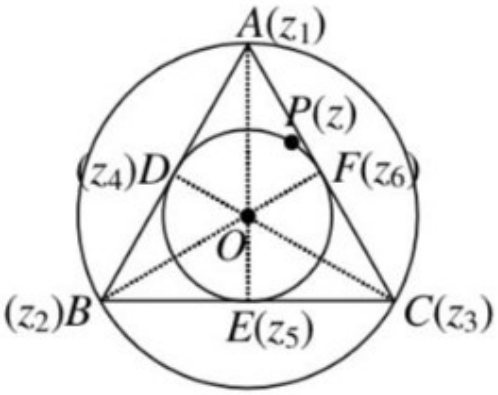
$\Rightarrow AB^2 = OA^2 + OB^2 - 2\,\mathrm{Re}\, z_1 - \bar{z}_2 \Rightarrow RE\left(z_1 - \bar{z}_2\right) = -2$

Similarly, $\mathrm{Re}\left(z_1\bar{z}_3\right) = \mathrm{Re}\left(z_3 - \bar{z}_1\right) = -2 \Rightarrow \mathrm{Re}\left(z_1\bar{z}_2 + z_2\bar{z}_3 + z_3\bar{z}_1\right) = -6$

B) Since $\angle AOC = \frac{2\pi}{3}$

$\therefore \frac{z_1 - 0}{z_3 - 0} = \frac{2}{2}e^{\frac{2\pi i}{3}}$

$\Rightarrow \frac{4z_1}{z_3} = \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) \times 4 = 2\left(-1 + i\sqrt{3}\right) \Rightarrow a = 2$



C) We have $|z_1 + z_2|^2 + |z_2 + z_3|^2 + |z_3 + z_1|^2$

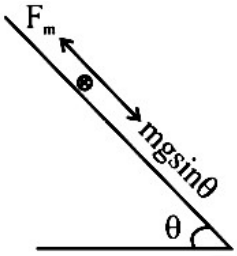
$$= 2\left(|z_1|^2 + |z_2|^2 + |z_3|^2\right) + 2\,\mathrm{Re}\left(|z_1\bar{z}_2| + |z_2\bar{z}_3| + |z_3\bar{z}_1|\right) = 12$$

D) $DP^2 + EP^2 + FP^2 = |z - z_4|^2 + |z - z_5|^2 + |z - z_6|^2 = 3|z|^2 + |z_4|^2 + |z_5|^2 + |z_6|^2 = 6$

For More Join: @IITJEE_Advanced

PHYSICS

17. $F_m = mg\sin\theta$



$ILB = mg\sin\theta \qquad \left(\frac{BLv\cos\theta}{R}\right)LB = mg\sin\theta \qquad v = \frac{mgR\tan\theta}{B^2L^2}$

18. Energy of light $E_1 = \frac{1240}{620} = 2eV \quad E_2 = \frac{1240}{496} = 2.5eV \quad E_3 = \frac{1240}{310} = 4eV$

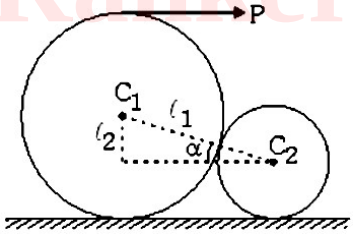
No. of photons capable to emit electrons per second

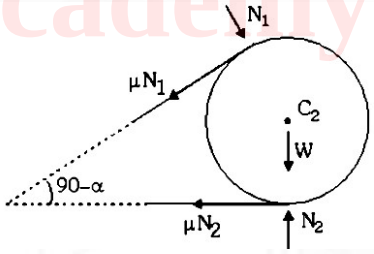
$$= \frac{\left(\frac{384}{3}\right)\times10^{-6}}{2.5e} + \frac{\left(\frac{384}{3}\right)\times10^{-6}}{4e} = \frac{83.2\times10^{-6}}{e}$$

No. of electrons emitted per second $= \frac{83.2\times10^{-6}}{2\times e}$

Photo current $= ne = 41.6\times10^{-6}$ ampere.

19. $\ell_1 = \frac{(D+d)}{2} \qquad \ell_2 = \frac{(D-d)}{2}$





$\sin\alpha = \left(\frac{D-d}{D+d}\right) \qquad \cos\alpha = \frac{2\sqrt{dD}}{(D+d)}$

Since smaller sphere remains at rest

$\mu N_1 d = \mu N_2 \qquad N_1 = N_2$

(Net torque about C_2 must be zero).

Net horizontal force on smaller sphere is zero.

$\mu N_2 + \mu N_1\left[\cos(90-\alpha)\right] = N_1\cos\alpha$

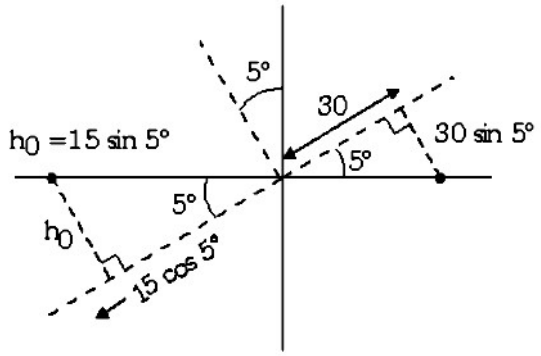
$\mu N_1 + \mu N_1\sin\alpha = N_1\cos\theta\,(N_1 = N_2)$

$\mu(1+\sin\alpha) = \cos\alpha \qquad \mu\left[1+\left(\frac{D-d}{D+d}\right)\right] = \frac{2\sqrt{Dd}}{(D+d)}$

$\mu = \sqrt{\frac{d}{D}} \qquad \mu \geq \sqrt{\frac{d}{D}}$

For More Join: @IITJEE_Advanced

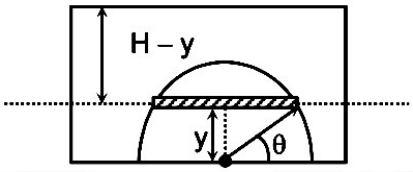
20.



$\frac{1}{v}-\frac{1}{-15}+\frac{1}{10}$ $\frac{1}{v}-\frac{3}{30}-\frac{2}{30}$

$V=30$ $m_2-2=\frac{h_i}{h_o}$ $h_i=30\sin 5^\circ$

21. $dF=pdA=\rho g\left(H-R\sin\theta\right).2R\cos\theta.R\cos\theta d\theta$



$$F=2R^2\rho g\int_0^{\pi/2}\left(H-R\sin\theta\right)\cos^2\theta d\theta=2R^2\rho g\left(\frac{H\pi}{4}-\frac{R}{3}\right)\ d\tau=dFy$$

$$\tau=2R^3\rho g\int_0^{\pi/2}\left(H-R\sin\theta\right)\cos^2\theta\sin\theta d\theta=2R^3\rho g\left(\frac{H}{3}-\frac{\pi R}{16}\right)$$

22. $\lambda=\frac{3500}{1-\frac{1}{p^2}}$ $\lambda_{max}=\frac{3500}{1-\frac{1}{4}}=466.67\,\text{nm}$

23. $\left|\frac{\Delta Y}{Y}\right|_{max}=\frac{\Delta m}{m}+\frac{\Delta L}{L}+\frac{2\Delta d}{d}+\frac{\Delta x}{x}$

$\frac{0.1}{30}+\frac{1}{200}+\frac{2(0.001)}{0.060}+\frac{0.001}{0.200}=0.0466$ $\left|\frac{\Delta Y}{Y}\right|_{max}=4.66\%$

24. Net elongation $=\int_0^L\frac{F_x}{A_xY_x}dx=\frac{F_0}{Y_0A_0}\int_0^L\left(1+\frac{x}{L}\right)dx=\frac{3F_0L}{2Y_0A_0}$

25. Let impulse be $\int Fdt$

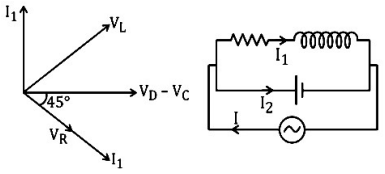
$\therefore\int\vec{F}dt=mv\hat{j}-mv\hat{i}$

Also, $\int\vec{F}dt=2m\vec{v}_f+2m\left(\frac{v}{2}\right)\hat{j}$

$\therefore\vec{v}_f=-\frac{v}{2}\hat{i}$ $\therefore\alpha=1$

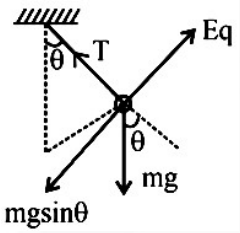
For More Join: @IITJEE_Advanced

26.



$$\tan\theta = \frac{I_2 \sin 135}{I_1 + I_2 \cos 135} = \frac{\frac{V_0}{8}\frac{1}{\sqrt{2}}}{\frac{V_0}{4\sqrt{2}} - \frac{V_0}{8\sqrt{2}}} \quad \therefore \tan\theta = 1 \Rightarrow \theta = 45^\circ$$

27. For equilibrium, $mg\sin\theta = Eq$



$$\therefore \sin\theta = \frac{Eq}{mg} \Rightarrow \theta = 30^\circ$$

28. $\phi = B_0 t^2 a^2 \qquad \Rightarrow \in = 2B_0 a^2 t \quad \Rightarrow i = \frac{2B_0 a^2 t}{r}$

$$M = \frac{2B_0 a^2 t}{r} \times a^2 \left(\hat{j}\right)$$

$$Z = M \times B = \frac{2B_0^2 a^4 t}{r} \text{ clocksize}$$

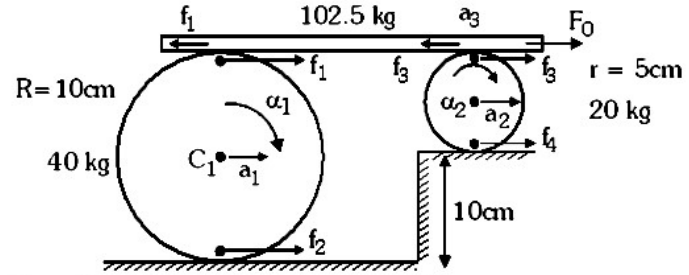
Limiting case: $\frac{2B_0^2 a^4 t}{r} = mg\frac{a}{2} \quad \Rightarrow t = \frac{mgr}{4B_0^2 a^3}$

29. $N_{max} = Mg + Ma_0 + M\omega^2 A \qquad = 960$ Newton

30. $\frac{L}{10} \Rightarrow \frac{\pi}{4}, \frac{L}{2} = \frac{5\pi}{4}, \lambda = \frac{v}{f} = \frac{1}{2}m$

$$L = \frac{5\pi}{4} \Rightarrow 8L = 8 \times \frac{5\pi}{4} = 5$$

31.



$F_0 - f_1 - f_3 = m_3 a_3 \qquad \ldots.(1)$

for plank

$f_1 + f_2 = m_1 a_1$

$f_1 R_1 - f_2 R_1 = \left(\frac{1}{2} m_1 R_1^2\right)\left(\frac{a_1}{R_1}\right)$

$f_1 - f_2 = \frac{40 a_1}{2}$

$f_1 - f_2 = 20 a_1$

$f_1 + f_2 = 40 a_1$

For More Join: @IITJEE_Advanced

$f_1 + f_2 = m_1a_1$ ....(2)

for solid bigger sphere

$f_1R_1 - f_2R_2 = I_{cm1}\alpha_1$ ....(3)

$a_1 = R_1\alpha_1$ ....(4)

$f_3 + f_4 = m_2a_2$ ....(5)

for smaller sphere

$f_3R_2 - f_4R_2 = I_{cm2}\alpha_2$ ....(6)

$a_2 = R_2\alpha_2$ ....(7)

$a_3 = 2a_1$ ....(8)

$a_3 = 2a_2$ ....(9)

$2f_1 = 60a_1,\ f_1 = 30a_1$

$f_2 = f_1 - 20a_1 = 10a_1$

$f_2 = 10a_1$

$f_3 + f_4 = 20a_2$

$f_3 - f_4 = \frac{1}{2}MR_2^2\frac{a^2}{R}$

$f_3 - f_4 = \frac{Ma_2}{2}$

$f_3 + f_4 = 20a_2$

$f_3 - f_4 = 10a_2$

$2f_3 = 30a_2$

$f_3 = 15a_2$

$f_4 = 20a_2 - f_3 = 20a_2 - 15a_2$

$f_4 = 5a_2$

$F_0 - f_1 - f_3 = (102.5)a_3$

$25 - 30a_1 - 15a_2 = (102.5)a_3$

$25 - 30 \times \frac{a_3}{2} - 15\left(\frac{a_3}{2}\right) = 102.5 \times a_3$

$25 - 15a_3 - 7.5a_3 = 102.5a_3$

$25 = 125a_3$

$a_3 = \frac{1}{5} = 0.2\ m/s^2$

$a_1 = 0.1$ $a_2 = 0.1$

$f_1 = 30 \times 0.1 = 3$ $f_3 = 1.5$

$f_2 = 10 \times 0.1 = 1$ $f_4 = 0.5$

32. Constants: $\lambda = 6000 = 600\,nm, D = 1.20\,m,$

$d = 1.2\,mm = 1.2 \times 10^{-3}\,m.$

(P) Net shift of central maximum when slabs t_1, μ_1 and t_2, μ_2 placed before S_1, S_2;

Optical path difference introduced,

$\Delta = (\mu_1 - 1)t_1 - (\mu_2 - 1)t_2.$

Shift on screen $x = \Delta\frac{D}{d}.$

Here $(1.6-1)0.02\,mm - (1.4-1)0.03mm = 0$

$\Rightarrow x = 0. \rightarrow 0\,nm.$

(Q) To make intensity at centre zero we need additional path difference

$\Delta = \frac{\lambda}{2}(\text{minimum}, n = 0).$

With slab of refractive index $\mu = 1.5$ on one slit,

$(\mu - 1)t = \lambda/2 \Rightarrow t = \frac{\lambda}{2(\mu-1)} = \frac{600nm}{2 \times 0.5} = 600\,nm.$

(R) Two equal sources: $I = 4I_0\cos^2(\delta/2).$

Given $I = 0.75\,I_{central} = 0.75 \times 4I_0 = 3I_0.$

So $\cos^2(\delta/2) = 3/4 \Rightarrow \delta/2 = \pi/6$

$\Rightarrow \delta = \pi/3.$ Path difference $\Delta = \frac{\delta}{2\pi}\lambda = \frac{\lambda}{6} = 600/6 = 100nm.$

(S) Fringe width $\beta = \frac{\lambda D}{d}.$ If screen moves right at 10 cm/s for 10s, $\Delta D = 1.0$ m.

Change in fringe width $\Delta\beta = \frac{\lambda\Delta D}{d} = \frac{600 \times 10^{-9} \times 1}{1.2 \times 10^{-3}} = 5 \times 10^{-4}\,m = 5 \times 10^5\,nm.$

Given $\Delta\beta = 1000x\,nm \Rightarrow 1000x = 500000 \Rightarrow x = 500$

For More Join: @IITJEE_Advanced

CHEMISTRY

33. NCERT

34. Conceptual

$$N_2(g)+3H_2(g)\rightleftharpoons 2NH_3(g)$$

35.

Initial moles	1	3	0
Moles of equilbrium	$1-x$	$3-3x$	$2x$

Total moles of gases = $(1-x)+(3-3x)+2x=4-2x$

Equilibrium partial pressure

$\frac{1-x}{4-2x}\times P$ $\frac{3-3x}{4-2x}\times P$ $\frac{2x}{4-2x}\times P$

Now, $K_P=\dfrac{\left(\frac{2x}{4-2x}\times P\right)^2}{\left(\frac{1-x}{4-2x}\times P\right)\left(\frac{3-3x}{4-2x}\times P\right)^3}$

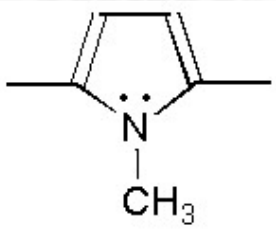
$=\dfrac{4x^2}{27(1-x)^4}\times\dfrac{(4-2x)^2}{P^2}=\dfrac{4x^2\times 16}{27\times P^2}$

$\therefore\ x=\sqrt{\dfrac{27K_P.P^2}{64}}=\dfrac{3P.\sqrt{3K_P}}{8}$

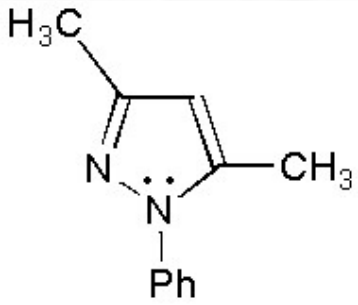
36. Conceptual

37.

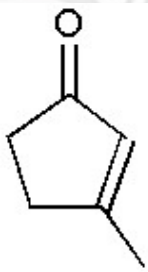
A)



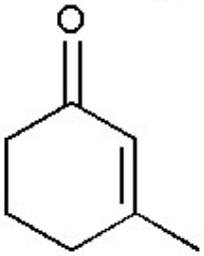
B)



C)



D)



38. Theory based.

39. NCERT

40. $p=5, q=7$

$\therefore\ p\times q=35$

Rankers Academy JEE

For More Join: @IITJEE_Advanced

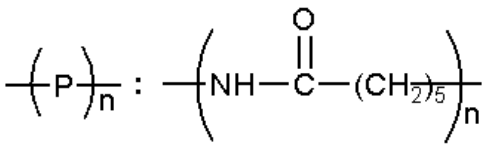
41. Except $\left[Ni(CO)_4\right];\left[Ni(CN)_4\right]^{2-};\left[NiF_6\right]^{2-}$ all others contain 2 unpaired electrons

42. NCERT P.no: 438 (II year), 391 (I year)

$X:(CH_2)_5(COOH)_2$

Y : Cyclohexanone

Z : Caprolactum



Formula weight of $P=113$

43. Only Toluene

$+I$ and $+H$ groups (alkyl) only activates all $o-,m-,p-$positions in AES.

44.

	$COCl_2 \longrightarrow$	CO	Cl_2
Initial pressure at $2T$	**40**	-	-
Equilibrium pressure	**20**	**20**	**20**

Co differs from V to $4V$.

$\therefore$ Its partial pressure becomes 5 mm of Hg.

$P_1 = 20+20+5 = 45\,\text{mm of Hg}$

45. $E = E^\circ - \frac{0.06}{n}.\log\frac{P_{H_2}}{\left[H^+\right]^2}$

$\left(\text{assuming } 2H^+ + 2e^- H_2\right)$

$-0.3 = 0 - \frac{0.06}{2}.\log\frac{1}{\left[H^+\right]^2} \Rightarrow \left[H^+\right] = 10^{-5}M$

$NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$

$K_h = \frac{\left[H^+\right]^2}{\left[NH_4^+\right]} = \frac{10^{-10}}{0.025} = 4\times10^{-9}$

For 0.1 M solution

$h = \sqrt{\frac{K_h}{C}} = \sqrt{\frac{4\times10^{-9}}{0.1}} = 2\times10^{-4}$

47. N < F < O < Li : Increasing order of IE_2

F < N < Si < Mg < Rb : Increasing order of radius

K < Na < Al < Mg : Increasing order of IE_1

N < B < O < Cl : Increasing order of $|EA|$

Rankers Academy JEE
Sri Chaitanya Educational Institutions INDIA
For More Join: @IITJEE_Advanced